

Scenario 02 – File Upload

I – Instruction

Generate comprehensive functional, negative, boundary, security, usability, and performance test cases for the File Upload feature.

C – Context

Users can upload PDF, DOCX, JPG, and PNG files. Supports single and multiple uploads. Min size 1 KB, max size 10 MB. Reject invalid formats, oversized, corrupted, duplicate, password-protected (if unsupported), and malicious files. Validate upload progress, retry, cancellation, and network interruption.

E – Example

A valid 5 MB PDF uploads successfully. Uploading an EXE or a 15 MB PDF displays an appropriate validation error.

P – Persona

Act as a Senior QA Automation Engineer specializing in web application security, OWASP validation, and file upload testing.

O – Output Format

Generate a table with Test Case ID, Category, Scenario, Preconditions, Input File, Steps, Expected Result, Priority, Severity, and Automation Feasibility.

T – Tone

Comprehensive, security-focused, risk-based, and industry-standard.

Observation

The original prompt lacks file constraints, validation rules, and security requirements. The ICEPOT prompt adds complete context, examples, and expected outcomes for comprehensive test coverage.

Scenario 04 – Role-Based Access Control

I – Instruction

Generate detailed functional, authorization, negative, and security test cases for Role-Based Access Control.

C – Context

Roles: Admin, Manager, Editor, Viewer. Validate permission matrix, unauthorized access, direct URL access, API authorization, privilege escalation, session changes, and audit logging.

E – Example

A Viewer attempting to delete a record should receive HTTP 403 Access Denied, and no data should be modified.

P – Persona

Act as a Senior QA Engineer specializing in RBAC, authorization, and OWASP security testing.

O – Output Format

Generate a permission matrix and a table with Test Case ID, Role, Module, Scenario, Steps, Expected Result, Severity, Priority, and Automation Scope.

T – Tone

Enterprise-grade, security-oriented, and compliance-focused.

Observation

The original prompt does not define roles, permissions, or access rules. The ICEPOT prompt specifies authorization scenarios and role hierarchy for accurate RBAC testing.

Scenario 05 – API Testing

I – Instruction

Generate comprehensive API test cases for the User Registration endpoint covering functional, validation, negative, security, performance, and contract testing.

C – Context

POST /api/v1/users/register. Validate request payload, response schema, status codes (201,400,401,403,409,422,429,500), authentication, duplicate users, SQL Injection, XSS, rate limiting, invalid JSON, response headers, and response time. Tools: Postman and Rest Assured.

E – Example

A valid request returns HTTP 201 with User ID. Existing email returns HTTP 409 Conflict.

P – Persona

Act as a Senior API Test Automation Engineer experienced in REST APIs, Postman, Rest Assured, OpenAPI validation, and CI/CD.

O – Output Format

Generate a table with Test Case ID, Category, Request Payload, Preconditions, Steps, Expected Status Code, Expected Response, Priority, Severity, and Automation Scope.

T – Tone

Detailed, API-first, enterprise-standard, and automation-ready.

Observation

The original prompt misses API specifications such as endpoint, payload, status codes, and validation rules. The ICEPOT prompt provides complete API context for automation-ready test cases.